Salifort Motors

Predicting Employee Attrition: A Strategic Data-Driven Framework for Long-Term Retention

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The Challenge: Understanding Churn

Attrition Costs

Employee turnover at Salifort Motors isn't just a HR metric—it's a significant financial and cultural burden. Replacing top talent involves high recruitment, training, and opportunity costs.

The Data Goal

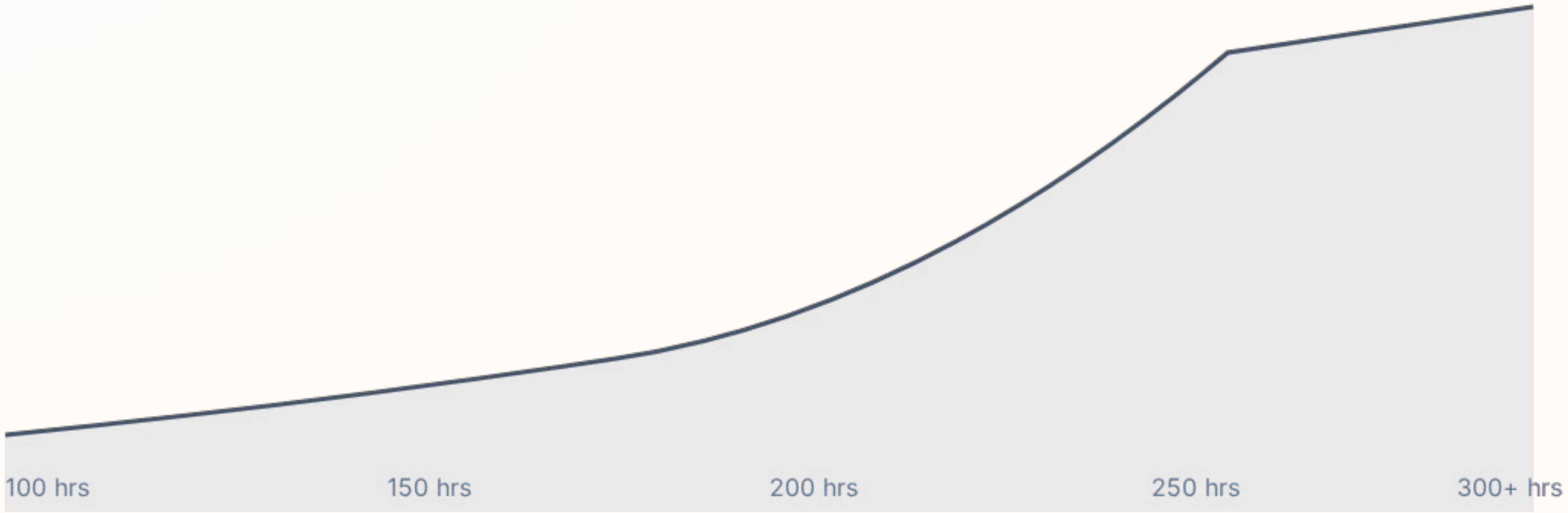
Our objective was to analyze historical employee data to identify the "tipping points" that lead to resignation, shifting HR from reactive responses to proactive intervention. In order to do this, an employee survey was conducted and results were tabulated.

Visualizing Attrition Patterns

Initial analysis revealed two distinct "high-risk" clusters of employees who left the company:

- **The Overworked:** Employees with 6-7 projects and 250+ average monthly hours.
- **The Disengaged:** Employees with low satisfaction and under-utilization (2 projects).

The Workload/Retention Paradox



Risk of attrition spikes exponentially after 250 average monthly hours.

The Analytical Approach

Diagnostic
Establishing baseline metrics using Logistic Regression and Naive Bayes to identify key statistical significance.

Stable
Building a pruned Random Forest to prioritize transparency and ease of interpretation for stakeholders.

Advanced
Deploying XGBoost to capture complex non-linear interactions between variables like tenure and satisfaction.

The Aggressive Challenger: XGBoost

0.97

Recall Score

Technical Trade-offs

While XGBoost achieved our highest recall, the model architecture is complex—utilizing 500 trees at a depth of 7. This "aggressive" approach risks overfitting and consumes significant computational resources, leading us to favor a more balanced solution.

The Selection: Random Forest

The **Random Forest** was chosen for deployment due to its exceptional stability and interpretability. It achieves a robust 0.92 recall and 0.98 accuracy without the excessive complexity of the boosted engine.

This "modest" choice ensures that Salifort's HR team can trust the feature importances when making critical personnel decisions.

Model Performance Comparison

Metric	Logistic Regression	Random Forest (Winner)	XGBoost
Accuracy	0.82	0.98	0.99
Precision	0.44	0.99	0.98
Recall (Leavers)	0.26	0.92	0.97
Complexity	Low	Moderate	High

Top Drivers of Attrition

Visualized using the "flare_r" inspired aesthetic for clarity and impact.

Strategic HR Recommendations

- **Workload Caps:** Implement soft caps on projects (max 5) and monthly hours (max 220) for tenured staff.
- **Pulse Surveys:** Deploy quarterly satisfaction checks to identify disengaged clusters.
- **Tenure Milestones:** Introduce specialized growth paths for employees at the 4-year mark.
- **Mentorship:** Pair high-eval performers with mentors to manage burnout risk.

Implementation Timeline

Q1 - Pilot

Deploy model to Sales and HR departments for testing.

Q2 - Integrate

Incorporate attrition risk scores into HR dashboard.

Q3 - Review

Compare predicted leavers vs actual turnover stats.

Q4 - Scale

Roll out across all Salifort global offices.

Questions?

Transforming Employee Data into Sustainable Retention

Project: Salifort Motors Capstone | Analytics Team